

Supervised Learning Models to Predict Firm Performance With Annual Reports: An Empirical Study

Xin Ying Qiu

*CISCO School of Informatics, Guangdong University of Foreign Studies, No. 178, Waihuandong Road, Higher Education Mega Center, Panyu District, Guangzhou, Guangdong, China, 510006.
E-mail: xinying.qiu@gmail.com*

Padmini Srinivasan

*Computer Science Department, 101F MacLean Hall, The University of Iowa, Iowa City, IA.
E-mail: padmini-srinivasan@uiowa.edu*

Yong Hu

Institute of Business Intelligence and Knowledge Discovery, Guangdong University of Foreign Studies, and School of Business, Sun Yat-sen University, No. 178, Waihuandong Road, Higher Education Mega Center, Panyu District, Guangzhou, Guangdong, China, 510006. E-mail: Henryhu200211@163.com

Text mining and machine learning methodologies have been applied toward knowledge discovery in several domains, such as biomedicine and business. Interestingly, in the business domain, the text mining and machine learning community has minimally explored company annual reports with their mandatory disclosures. In this study, we explore the question “How can annual reports be used to predict *change* in company performance from one year to the next?” from a text mining perspective. Our article contributes a systematic study of the potential of company mandatory disclosures using a computational viewpoint in the following aspects: (a) We characterize our research problem along distinct dimensions to gain a reasonably comprehensive understanding of the capacity of supervised learning methods in predicting change in company performance using annual reports, and (b) our findings from unbiased systematic experiments provide further evidence about the economic incentives faced by analysts in their stock recommendations and speculations on analysts having access to more information in producing earnings forecast.

Introduction

Predicting company financial performance has been an important research question in the business domain. Both numerical and textual data have been explored extensively to build forecasting models for company financial

performance. In the realm of relating textual corpora with financial performance, we have seen more research focusing on predicting intraday stock market performance with news articles (Mittermayer, 2004; Schumaker & Chen 2009; Tetlock, Saar-Tsechansky, & Macskassy, 2008), with message board sentiments (Das & Chen, 2007), or by combining news with short-term price indicators for prediction (Robertson, Geva, & Wolf, 2007; Zhai, Hsu, & Halgamuge, 2007). These forecasting models are usually designed to predict stock market volatility and responses within hours or even minutes of news events; however, company performance is not necessarily only evaluated within such a short time window. Investors and shareholders are often interested in information that could assist in their long-term investment decisions and in constructing investment portfolios.

Narrative discussions in mandatory information disclosures are widely acknowledged as important for evaluating firm performance and value; however, the machine learning and text mining community has rarely explored these texts. Such discussions have a purpose similar to those of the widely available financial data on a company's status: broadly, to disclose and disseminate information about company performance and future prospects and to maintain market efficiency and integrity. We believe that computational methods to mine the mandatory information disclosures in annual reports will be of value to investors when they are evaluating the potential long-term change in company financial performance.

Some relevant works in this direction are those of Magnusson et al. (2005), F. Li (2008, 2010), Feldman,

Received December 15, 2012; revised March 20, 2013; accepted April 2, 2013

© 2013 ASIS&T • Published online 20 November 2013 in Wiley Online Library (wileyonlinelibrary.com). DOI: 10.1002/asi.22983

Govindaraj, Livnat, and Segal (2010), and Lehavy, Li, and Merkley (2011). In Magnusson et al. (2005), company quarterly reports and corresponding financial ratios were clustered with prototype matching clustering and self-organizing maps clustering, respectively. Although the two sets of clusters did not coincide, the authors found that the changes in textual report patterns tend to precede and indicate the changes in financial performance. F. Li (2008) used the Fog index (Gunning, 1952) to measure the readability of annual reports. He found that firms with hard-to-read annual reports have lower earnings. Lehavy et al. (2011) further studied the linear association between the readability of annual reports and several measures of analyst behavior using regressions. In F. Li (2010), a naïve Bayesian machine learning algorithm was applied to study how the information contained in the forward-looking statements in annual reports are related to different financial indicators. Feldman et al. (2010) used regression analysis to show that tone changes in annual reports are associated with immediate market reactions and can be used to predict future stock prices. In general, these studies focused on specific features of company reports, such as readability and tone. We have not yet seen a systematic study exploring the predictive power of the narrative portions of the annual reports. Moreover, in the existing studies, models based on machine learning have not been compared against the predictions of analysts. These observations motivated our work presented in this article.

Our overall goal is to empirically and comprehensively assess the power of annual reports in predicting change in company financial performance. Here, “change” is a temporal notion that is measured by comparing performance

over consecutive time periods, specifically over consecutive years for the same company. Due to the once-a-year release interval of annual reports, we hypothesize that annual reports contain textual information relevant to investors’ decisions for the following year. Reflecting real-life practice, we assume that models built with data of a given year will be useful to forecast new data in the following year. We propose to address our research goal with predictive models built by selecting options along five dimensions of a strategy based on text mining and designed particularly for annual reports prediction: type of learning model, financial performance indicators, evaluation measures, document representation model, and experiment design. The five-dimensional research framework that guides our experiments is presented in Figure 1.

We consider various options along each dimension. With the experiment design dimension, models built with a given year’s data are used to forecast in the following year. We explore two temporal measures of firm performance change based on earnings per share (EPS) and size-adjusted cumulative return (SAR). With each measure, we study a three-class classification problem in which a firm needs to be categorized as outperforming, average, or underperforming. We also apply a support vector machine (SVM) based regression model to tap into the predictive potential of annual reports. Model performance is measured using, for example, standard accuracy, costs of errors, Kendall’s τ , and Spearman’s ρ as well as by comparisons against analysts’ forecasts. We further analyze our predictive models by replicating experiments over different years. By exploring the different options and their combinations along

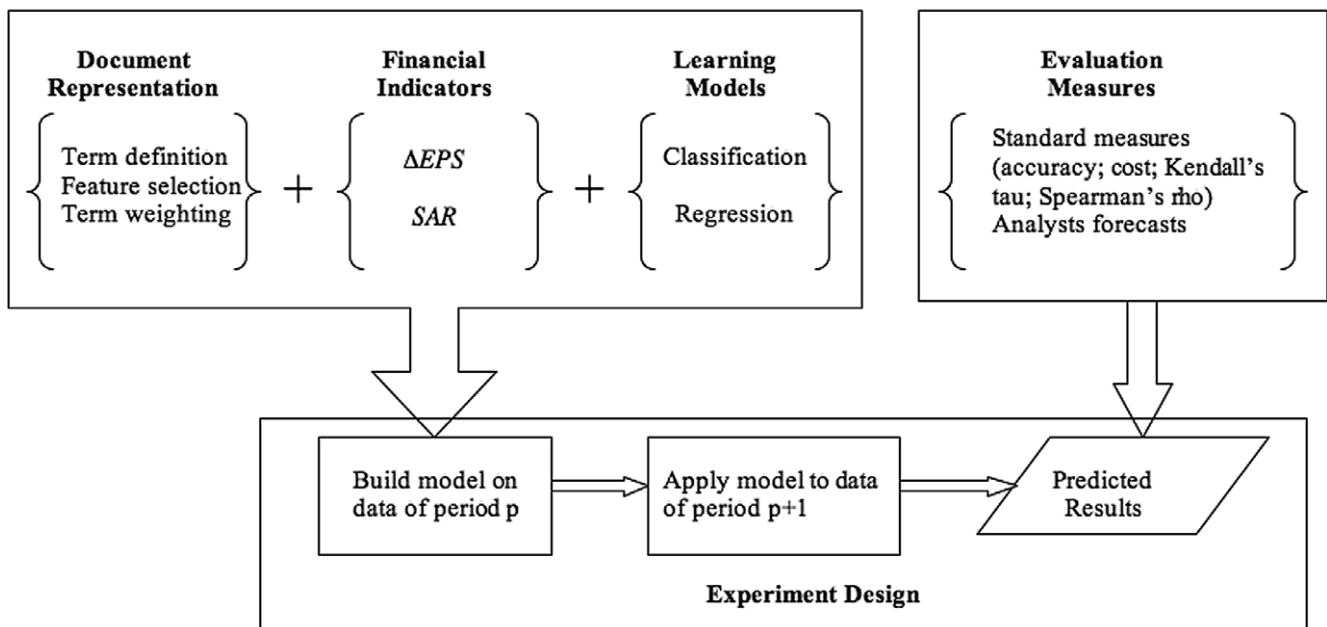


FIG. 1. Research framework composed of five dimensions: document representation, financial indicators, learning models, evaluation measures, and experiment design. (Terms in parentheses are options to consider within each dimension; terms in the “Document Representation” dimension are subdimensions to consider with options omitted to maintain figure simplicity.)

the five dimensions of our framework, we aim toward a comprehensive understanding about the predictive capacity of text mining models using annual reports. Specific research questions that we address are the following:

RQ1: What are the relative merits of the different classifiers when applied to our goal? (Here, we compare the different SVM-based classifier models.)

RQ2: How do our different supervised learning models perform compared to analysts' forecasts? (Here, we test our best SVM-based classifier and our regression model.)

RQ3: Are annual reports suitable for predicting change in certain financial measures? (Here, we compare EPS and SAR.)

RQ4: How do our predictive models perform over different years? (Here, we compare experiment results with data around the financially turbulent year of 2000.)

These questions that are at the heart of our research are designed to inform us about the extent to which annual reports may be exploited to predict change in performance. Our five-dimensional research framework allows us to systematically test different text mining strategies by varying document representation, learning model, design, and so on (detailed later). In this way, we provide the groundwork for applying supervised learning models to business document mining.

The major design, results, and analysis of this article come from a portion of the first author's PhD thesis completed in 2007. However, significant recent revisions and extensions are included: (a) We present results using a better term-weighting scheme in document representation; (b) the results with the regression models are new (this includes evaluation with Kendall's τ and Spearman's ρ), and the regression results provide a more comprehensive analysis and discussion of the research questions; and (c) we also updated our review of related research and that provides further validation and foundation for this work.

Our results indicate, for example, that for some measures, analysts are more accurate than are our models, whereas this is not so for other measures. At a high level, our results are consistent with observations in the accounting/business literature. We found particular SVM models to be the strongest. Our regression results, though interesting, are on the whole not as strong as our classifier results. Overall, our article contributes a foundation in terms of both methodology and results for applying supervised learning models to predicting firm performance from their annual reports. The rest of the article is organized as follows: First, we present related research against our empirical framework. Next, we analyze our results addressing the four research questions raised earlier. Our conclusions and directions for future research are then presented.

Empirical Research Framework and Related Research

Our empirical framework is constructed along five dimensions: learning model, financial indicators, experiment

design, document representation models, and evaluation measure. We discuss our design and related research in this section. Given the vast amount of research in these areas, our review will be brief.

Learning Model

Our hypothesis is that the textual content of annual reports contains information that may be used to predict change in financial performance in the following year. We test this hypothesis using a *supervised learning method*, for which we pair annual reports in a given year with the corresponding companies' *change* in financial performance in the following year to form training data. We train a learning model using these data and then evaluate the model on unseen annual reports to test if the model can predict companies' change in financial performance. In the domain of supervised learning study, research questions can be categorized roughly into classification, tagging, and regression problems. We leave out the tagging model because it is a structure-prediction problem irrelevant to our research goal.

Text classification. Machine learning-based text-classification algorithms (Sebastiani, 2002) have been applied to problems in various domains. In biomedicine (for a comprehensive survey, see Shatkay & Feldman [2003]), text classification has been used to explore gene document annotation with ontology terms (Srinivasan & Qiu, 2007), to identify specific semantic relations between protein and subcellular structures (Rice, Nenadic, & Stapley, 2005), and to classify sentence segments along multiple annotation dimensions (Shatkay, Pan, Rzhetsky, & Wilbur, 2008). In the business domain, Y. Li and Bontcheva (2008) adapted the standard SVM algorithm with new techniques for multiclass, multilabel patent classification with multifaceted hierarchical categories.

Among the many classification algorithms applied to text classification, SVMs have been recognized as being able to efficiently handle text classification problems with many thousands of support vectors (Joachims, 1998). SVMs find a separating hyperplane in the high-dimensional space transformed from the original feature space through kernel functions such that this hyperplane achieves the maximum margin in terms of the distance between the data points and the hyperplane. The data points thus separated are classified into different classes. Different kernel functions augment the feature space with different information to determine the distance between data points. The basic kernel functions used in most SVM implementations are linear, polynomial, radial basis, and sigmoid. Previous research has shown that SVMs perform well when compared with classifiers such as naïve Bayes, Rocchio, and K-NN (Joachims, 1998; Kotsiantis, 2007). In our pilot study (Qiu, Srinivasan, & Street, 2006), we used the linear kernel function for SVM classification and achieved accuracy that was significantly better than baseline. We decide to continue exploring different models based on SVMs.

Regression. In predicting change in company performance, one reasonable learning model to consider is a regression model such as SVM regression (Smola & Scholkoph, 2004). The SVM regression model takes a training data set of vectors (in our case, annual report document vectors) and their associate target values (the financial measures), and fits a linear function that approximates the true relation between the vectors and the target real values. Regression has been applied to predicting financial performance with various financial reports' specific document features such as readability (Lehavy et al., 2011) and tone change (Feldman et al. 2010), but not with a comprehensive vector representation of the financial reports encompassing all term features, which is our plan. Because a company's financial performance each year has a natural ordering relative to its peers, another learning model to consider is ordinal regression (Chu & Keerthi, 2007), which solves the supervised learning problems of predicting target value of an ordinal scale. Ordinal regression bridges the approaches of metric regression and classification. In our research framework and experiments, we test both SVM regression and SVM ordinal regression as implemented in LibSVM. The results from SVM regression were consistent with those of the classification model, whereas SVM ordinal regression results were not encouraging when compared with those of the classification learning model. Although we plan on further analyzing these differences in another study, given these weaker results with the ordinal regression models when applied to our financial reports, we present only SVM regression results and analysis in this article.

Financial Performance: Measures and Indicators of Change

In the accounting and finance domains, there are many criteria for measuring firm performance. These include accounting measures such as operating earnings and market response measures such as stock return (Saad, Prokhorov, & Wunsch, 1998). Zhang, Cao, and Schniederjans (2004) compared neural networks and a variety of linear statistical models for forecasting EPS. EPS measures the amount of earnings per share of stock and serves as an indicator of a firm's profitability. Smith and Taffler (2000) employed a U.K. accounting ratio-based *z*-score to define company failure. Kloptchenko et al. (2004) selected seven ratios to characterize a firm's financial performance; these included three profitability ratios, one liquidity ratio, two solvency ratios, and one efficiency ratio. Thus, there is a variety of options for assessing firm performance. From the accounting research perspective, how to measure a firm's performance is fundamentally a question about accounting choices and their consequences (Fields, Lys, & Vincent, 2001). In this study, we consider both the accounting measure (i.e., EPS) and the market response measure (i.e., SAR) to evaluate a firm's financial strength and liquidity.

We denote year $t + 1$ to be the year for which we want to predict a company's change in performance, and year t to be

the given year of which the annual reports are used for prediction. Our set of annual reports and financial data cover 1,519 firms. (Details of data collection and description are provided later.) We use f_i to denote a single firm where $i \in [1, n]$, where n is the total number of firms. $EPS_t^{f_i}$, total earnings divided by the number of shares outstanding, shows how much of a firm f_i 's earnings at time t are available for distribution as dividends to each share of common stock. It helps investors decide on the potential of future dividends and the firm's ability to finance its growth internally. We denote a firm's EPS growth rate at year $t + 1$ relative to year t as: $\Delta EPS_{(t,t+1)}^{f_i} = (EPS_{t+1}^{f_i} - EPS_t^{f_i}) / |EPS_t^{f_i}|$. This definition is what we refer to as change in performance with respect to EPS.

SAR is inherently temporal. $SAR_{(t+1,t+2)}^{f_i}$ is the cumulative return from April of year $t + 1$ to March of year $t + 2$, minus the return of the corresponding market decile over the same period. The decile adjustment, which removes the average return for all firms with similar size, accounts for the market risk from investing in the sample firm. The SAR tells us the incremental return we may expect in a year if we invest in the firm at the end of March of year $t + 1$, based on the predictions made for year $t + 1$ (Balakrishnan, Qiu, & Srinivasan, 2010).

Class definitions. We use the characteristics of the distribution of actual firm performance under each measure to define classes. Figure 2 shows an example with $\Delta EPS_{(t,t+1)}^{f_i}$ for $t = 2002$. We see that the distribution is nearly normal. This observation holds for data of other years as well as for $SAR_{(t+1,t+2)}^{f_i}$. Given the near normal distribution for a particular year/measure combination, we categorize the top-25% firms as outperforming, the middle 50% as average-performing, and the bottom 25% as underperforming. With these three true classes of firms (determined for each year and measured using our historical data), the goal is to correctly classify each firm using predictive models built from prior-year annual reports.

As a note to our class definition, although our choice of 25–50–25% may seem to be arbitrary, it was decided based on our observations of the near normal distribution of the data. Additional validation for such a three-class decision is elaborated in Balakrishnan et al. (2010). To further investigate different class segmentation thresholds, Qiu (2010) employed a ranking method to vary the percentage of firms in each of the top, middle, and bottom classes. Our results indicate that the general trend in performance is consistent with the fixed percentage of 25–50–25% definitions. Thus, in this article, we only present results with the current categories of 25–50–25%.

Experiment Design

We ran experiments using a design that simulates the real-world application of our predictive model. In our design, the model is built with reports from one year (say, year $t - 1$), but tested with reports from the immediately

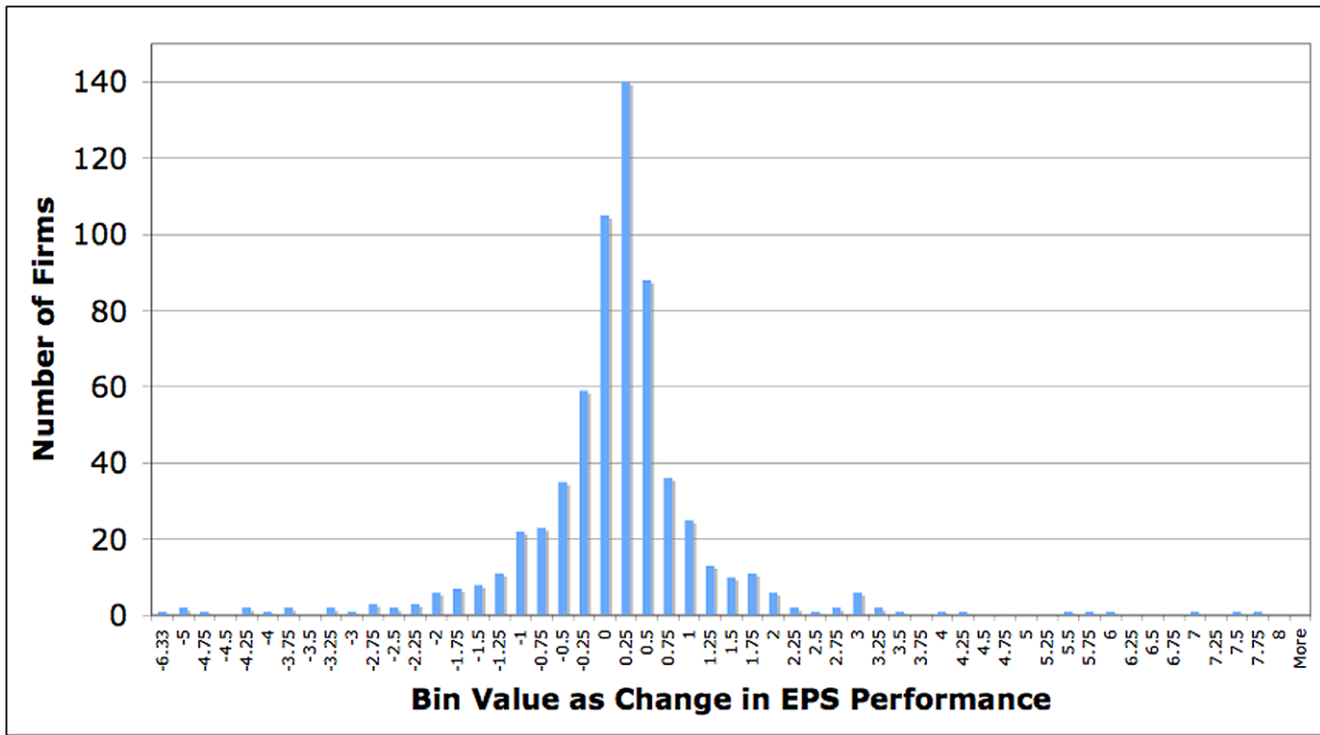


FIG. 2. Data histogram of change in firm performance (i.e., $\Delta EPS_{(t,t+1)}^{f_i}$) for Year 2002. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

following year (year t). Figure 3 shows the experiment design in pseudocode. The training data lag behind the test data by 1 year. With EPS for example, $\Delta EPS_{(t-1,t)}^{f_i}$ is used for training and $\Delta EPS_{(t,t+1)}^{f_i}$ for testing. Here $\Delta EPS_{(t-1,t)}^{f_i}$ represents change in EPS in year t compared to $t-1$ for firm f_i . $SAR_{(t+1,t+2)}^{f_i}$ tells us the incremental return we may expect in a year if we invest in the firm at the end of March of year $t+1$. We seek to predict financial performance in year $t+1$ using annual reports of year t that are made available in March of year $t+1$. All firm reports of a given year are used for training. The single model built is applied to all reports of the following year to categorize firms into three classes, specifically to predict firms' SAR or change in EPS of the following year. We evaluate our predictive model's performance with average accuracy and average cost of errors for the classification model, and Kendall's τ and Spearman's ρ for the regression model. In Steps 3.2 and 3.3 of Figure 3, we also compare performance of our models with the performance of analysts via their forecasts. The calculations for the analyst forecasts are described later.

SVMs are designed mainly to solve binary classification problems. Because we have a three-class classification problem, we need to consider different options. First, we could perform a one-against-rest classification for each class and combine the results to make a final decision. Second, we could perform a one-against-one classification for $n(n-1)/2$ pairs of classes and combine the results to make a final

decision. Third, we could use algorithms designed specifically for multiclass classification. The disadvantage of the second option is that it requires the largest number of classifiers, which in turn depends on the number of classes n . Thus, we decided to use Options 1 and 3 for our current study.

For Option 1, we use linear SVM to produce three one-against-rest models. There are two variants of this in terms of how we combine the results of the three models. First, because we use three binary classifiers to predict the three classes of outperforming, average, and underperforming, each firm will have three scores assigned to it by each of the three classifiers. Our first strategy for combining is to use the highest score to assign a class label. We denote this model as *SVM-score*. Second, we use Lin-Platt's method (Lin, Lin, & Weng, 2007; Platt, 1999) to transform each of the three scores into a probability that the firm belongs to one of the three classes. Then, we use the highest probability to assign a class label to the firm. We denote this model as *SVM-prob*. Our third algorithm is one designed specifically for multiclass classification, specifically the package implemented by Joachims based on Crammer and Singer (2002) study. We denote this model as *SVM-multi*.

For the regression model, we employ the LibSVM implementation of the SVM regression algorithm (Smola & Scholkoph, 2004), which has been applied successfully to various domains such as face detection (Y. Li, Gond, & Liddell, 2000) and travel-time prediction (Wu, Ho, & Lee,

```

1. Input:
  1.1 t: year in {1997, 1998, ..., 2003}
  1.2 m(f,t): change in firm f's performance from year t to t+1 using measure m
  1.3 r(f,t): annual report for firm f issued in year t
  1.4 F: set of firms
  1.5 M(t): {m(f,t)} for firms f in F
  1.6 R(t): {r(f,t)} for firms f in F
  1.7 C: 3 classes of firms: out-performing, average, and under-performing
  1.8 Pred-m(f,t): predicted change in firm f's performance from year t to t+1 using measure m and regression model

2. Process:
  2.1 Given F, M(t), M(t-1), R(t) and R(t-1)
  2.2 G-training = Determine-True-Classes(F, M(t-1)) (Section II.2)
  2.3 G-testing = Determine-True-Classes(F, M(t))
  2.4 Learning with classification model
    2.4.1 Build classifier using G-training and corresponding R(t-1) where f in F
    2.4.2 Apply classifier to reports R(t) where f in F
  2.5 Learning with regression model
    2.5.1 Build regression model using G-training and corresponding R(t-1) where f in F
    2.5.2 Apply regression to report R(t) where f in F

3. Output:
  3.1 Output from classifiers:
    O: {(f,c)}: Each f in F is classified into a class c in C
  3.2 Output from regression model:
    O: {(f, Pred-m(f,t))}: Each f in F is assigned a predicted measure m value for year t
  3.3 Results from classifiers = Compute accuracy and cost of errors of O with G-testing
    as gold standard
  3.4 Results from regression = Compute Kendall's tau and Spearman's rho with M(t) as gold standard

3. Evaluation:
  3.1 Compare classifiers results in terms of accuracy and cost
  3.2 Compare best classifier's accuracy with analysts accuracy for each financial measure m
  3.3 Compare regression model's Kendall's tau and Spearman's rho with those of analysts for each financial measure m.
  3.4 Analyze predictive performance by year and by financial measure

```

FIG. 3. Experiment design pseudocode.

2004). With SVM regression (SVR), we want to find a function $f(x)$ that approximates all training data (x_i, y_i) while tolerating no more than ε error from the target value of y_i . The linear case of such a function $f(x)$ needs to be as flat as possible, implying a small weight vector w . Slack variables ξ_i, ξ_i^* are introduced for feasible dual problem formulation. The constant C is introduced to balance the trade-off between the flatness of the function and the amount of deviation tolerable over ε . Formally, SVR can be formulated as follows:

$$\begin{aligned}
 & \min \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n (\xi_i + \xi_i^*) \\
 & \text{s.t. } y_i - (w \cdot x_i + b) \leq \varepsilon + \xi_i \\
 & \quad (w \cdot x_i + b) - y_i \leq \varepsilon + \xi_i^* \\
 & \quad \xi_i, \xi_i^* \geq 0
 \end{aligned}$$

where w is the weight vector in the function $f(x) = \langle w, x \rangle + b$, x is the data vector, $b \in R$, y is the target vector, C is

the cost parameter for the trade-off between flatness and error-tolerance, ε is the error, and ξ and ξ^* are the slack variables.

Document Representation

Document representation is in the form of a vector of weighted term features. This requires the following decisions: What is the definition of a term feature? How does one select the best subset of term features? How does one weight the selected features? For term feature definition, we use a stemmed, bag of words approach because it is the most widely adopted approach in document representation. More complex term definition methods such as phrases and n-grams have not been proven more definitively effective than is “bag of words” in text classification (Moschitti & Basili, 2004). We also considered the options of employing terms or phrases from domain-specific or general-purpose dictionaries to construct our document vector representation. However, F. Li (2010) found from systematic experimentation that commonly used dictionaries may not work well for analyzing corporate filings. Thus, we decided to use stemmed terms as our features.

For feature selection, our earlier work (Qiu et al., 2006) from testing multiple feature selection measures shows that document frequency (DF) with a 1% of training documents’ threshold offers optimal reduction of feature set size without increasing the cost of classification accuracy. We used this strategy here as well.

To select the best weighting schemes for feature vectors, we also explored different tf-idf term weighting schemes to construct the document vectors (i.e., “lfc,” “atc,” and “atn”; Manning & Raghavan, 2008). Our experimentation in the original thesis compared the performance of “lfc” and “atc,” and adopted an “atc” weighting scheme because it performs similarly as “lfc.” Later (Balakrishnan et al., 2010), we found that an “atn” weighting scheme is more effective than are “lfc” and “atc.” We rerun all experiments with the “atn” weighting scheme defined as:

$$atn: w_i = \left(0.5 + 0.5 \times \frac{tf}{\max tf} \right) \times \ln \left(\frac{N}{n} \right)$$

where tf is raw term frequency, $\max tf$ is the highest term frequency in the document, N is the number of documents in the collection, n is the number of documents containing term i , and w_i is the weight of term i .

Evaluation Measures

Evaluating classifiers. Our classification models predict each firm/year data point as outperforming, average-performing, or underperforming. These may be compared against the true labels to determine accuracy. Accuracy rate (or $1 - \text{error rate}$) is the proportion of correctly classified samples of all samples. Given our 25–50–25% three-class definition, the majority-vote baseline benchmark is to declare all firms as average-performing. This has a 50%

accuracy. We also compared accuracies obtained by our models against accuracies calculated for analysts’ forecasts. This comparison with analysts’ forecasts offers an important perspective on performance. Next, we show how accuracies are calculated for analyst forecasts.

Accuracy assessments do not consider the kinds of classification errors made and their associated costs. Some errors can be more severe than others. There are two types of errors a model may make: One is to predict an outperforming (or underperforming) firm as underperforming (or outperforming), and the other is to predict an outperforming (or underperforming) firm as average, or to predict an average firm as outperforming or underperforming. The former error should have a higher cost because it is a misclassification over two levels, whereas the latter is over one level. Formally, we define cost as:

$$Cost = \begin{cases} 2 \times C, & \text{if } \begin{cases} \text{TrueClass} = \text{out} - \text{perform}, \text{ and} \\ \text{PredictedClass} = \text{under} - \text{perform or} \\ \text{TrueClass} = \text{under} - \text{perform}, \text{ and} \\ \text{PredictedClass} = \text{out} - \text{perform} \end{cases} \\ C, & \text{otherwise} \end{cases}$$

Analysts’ forecast benchmark Forecast on EPS. We denote the analysts’ EPS forecast for year t as $AnalystEPS_t^{fi}$. A firm’s actual EPS in year t was defined before as EPS_t^{fi} . The analysts’ predicted growth rate of EPS at year $t + 1$ relative to year t is defined as:

$$\Delta AnalystsEPS_{(t,t+1)}^{fi} = (AnalystEPS_{t+1}^{fi} - EPS_t^{fi}) / |EPS_t^{fi}|$$

Using a symmetric procedure to the one used to categorize $EPS_{(t,t+1)}^{fi}$, we first rank the firm/year data according to $\Delta AnalystsEPS_{(t,t+1)}^{fi}$. We then categorize the top 25% as analyst forecasted outperforming, the middle 50% as forecasted average-performing, and the bottom 25% as underperforming. We calculate the accuracy of analysts’ forecasts and compare this with the accuracy of our predictive models.

Forecast on SAR We use analysts’ stock recommendations (buy, sell, keep) as another baseline for our predictive model with SAR. Specifically, we use the consensus mean recommendations for year $t + 1$ outstanding as of the end of the fourth month after year t ’s fiscal year end (i.e., in April of year $t + 1$). These are scaled from 1 (*strong buy*) to 5 (*strong sell*). We group these into three recommendation classes based on the distribution of the recommendation scores. We denote the class of analysts’ recommendation for company f in year $t + 1$ as $AnalystsClassStock_{t+1}^{fi}$. Given $AnalystClassStock_{t+1}^{fi}$ for a set of companies and their true classes based on $SAR_{(t+1,t+2)}^{fi}$, we can calculate predictive accuracy of analyst forecasts for year $t + 1$. This is another benchmark for our SAR-focused text-classification model.

Evaluating regression models. The standard approaches to evaluating regression models such as mean absolute error and mean square error (Steel & Torrie, 1960) generally represent

the “true” cost of the prediction errors. In our study of building predictive models of annual reports, ranking companies’ performances can be the underlying goal in real-world applications, such as constructing investment portfolios (Qiu, 2010). The predicted ordering of the companies is of at least equal interest to investors in practice. In the information retrieval domain, widely used ranking-evaluation measures such as F-score require definition of “relevant” documents according to users’ information need or judgment. In our study, however, defining “relevant” documents (or “positive” companies with annual reports as surrogate) to formulate ranking measures will not be applicable while we still need to evaluate firms’ performances in reference to each other. Rosset, Perlich, and Zadrozny (2007) suggested and validated the use of modified Kendall’s $\hat{\tau}$ and Spearman’s $\hat{\rho}$ as a ranking-based evaluation measure for regression models. We chose to implement these two measures to evaluate our regression models.

Formally, following Rosset et al.’s (2007) notation, modified Kendall’s $\hat{\tau}$ and Spearman’s $\hat{\rho}$ are defined as:

$$\hat{\tau} = 1 - \frac{4T}{n(n-1)}; \quad \hat{\rho} = 1 - \frac{12R}{n(n-1)(n+1)}$$

where $\hat{\tau}$ and $\hat{\rho}$ indicate the nonparametric correlation based on sample quantities, T is the number of ranking order switches such that $T = \sum_{i < j} 1\{s_i > s_j\}$, R is the sum of weighted order switches such that: $R = \sum_{i < j} (j-i) \cdot 1\{s_i > s_j\}$, and S_i is the rank of observation i in the original order such that $s_i = |\{j \leq n : y_i \leq y_j\}|$.

Results and Analysis

Data Collection and Descriptive Statistics

We restricted our sample of firms to manufacturing industry (Standard Industrial Classification codes: 2000–3999) in the United States, having December as the month defining the end of fiscal year. We also restricted our experiments to data from 1997 to 2003. These restrictions make our experiments tractable and also ensure some degree of sample homogeneity. Our research goal is to predict change in financial performance using the previous year’s annual reports. Based on our definition of the two measures ($\Delta EPS_{(t,t+1)}^f$ and $SAR_{(t+1,t+2)}^f$), we need to collect the corresponding financial ratios.

For the experiments with $\Delta EPS_{(t,t+1)}^f$, and for each firm/year, we retrieve from the Institutional Brokers’ Estimate System (I/B/E/S)¹ database actual EPS, and analyst consensus mean for EPS forecast made in April of the fiscal year

TABLE 1. Distribution of documents used for experiments.

Year	No. of documents		
	Total	$\Delta EPS_{(t,t+1)}^f$	$SAR_{(t+1,t+2)}^f$
		Experiments	Experiments
1997	782	–	782
1998	857	746	765
1999	804	714	719
2000	771	663	726
2001	798	649	758
2002	743	647	710
2003	666	658	–
Total	5,421	4,104	4,460

Note. For each year, number of documents in the “Total” column is the union of documents in the third and fourth columns of documents used for the $\Delta EPS_{(t,t+1)}^f$ and $SAR_{(t+1,t+2)}^f$ experiments, respectively.

with forecast ending period in December of the fiscal year. The retrieved data range from 1997 to 2003, giving us $\Delta EPS_{(t,t+1)}^f$ data from 1998 to 2003.

For the experiments with $SAR_{(t+1,t+2)}^f$, we retrieve from the CRSP¹ database the monthly return and decile monthly return for each firm/year. We calculate size-adjusted cumulative return as the size-adjusted buy-and-hold return cumulated for 12 months from April 1 of the fiscal year to the next April. The SAR data range from 1997 to 2003.

The data collected for the aforementioned two financial measures give us a total of 1,809 firms. Next, we retrieve the annual reports for these 1,809 firms by first manually downloading the accession codes for these firms’ annual reports from MergentOnline.² The accession codes are the unique identifiers of the annual reports. We then use these accession codes to automatically retrieve the reports from the Edgar-Scan³ database. Of the 1,809 firms with financial data, we were able to retrieve at least one annual report for 1,519 firms. In total, we have 12,564 annual reports from 1,519 firms for 1996 to 2004.

There are 10 different submission types for annual reports, including 10K (10K filing), 10K405 (10K filing where regulation S-K Item 405 box is checked), 10KSB (10K filing for small business), and 10KT (10K transition report). We focus on 10K and 10K405 filings. Our final useable documents with matching financial performance measure values are 5,421 annual reports published between 1997 to 2003 corresponding to 1,276 firms. Table 1 identifies the number of documents used for each experiment. The numbers vary because each experiment (by year and measure) depends on the availability of the corresponding performance data. Thus, we see that certain experiments could be run only for particular years.

On the whole, the data-collection step involved nontrivial effort, including several cross-checking steps to catch errors. These data are available from the first author.

¹I/B/E/S and Center for Research in Security Prices (CRSP) database access can be obtained through Wharton Research Data Service at <http://wrds-web.wharton.upenn.edu/wrds/>

²<http://www.mergentonline.com>

³<http://www.sec.gov/edgar.shtml>

TABLE 2. Relative performance of SVM-score, SVM-prob, and SVM-multi classifiers.

Measure \ Model	Performance for $\Delta EPS_{(t,t+1)}^f$			Performance for $SAR_{(t+1,t+2)}^f$		
	SVM-score	SVM-prob	SVM-multi	SVM-score	SVM-prob	SVM-multi
Average rank (M accuracy, variance)	2.6 (0.4872, <0.001)	1.6 (0.5112, <0.001)	1.8 (0.5023, <0.001)	3 (0.4266, 0.003)	1 (0.5064, <0.001)	2 (0.4715, 0.001)
Average rank (M cost, variance)	2.8 (409.6, 425.3)	1.8 (366.8, 755.7)	1.4 (365.4, 588.3)	3 (471.4, 6886.3)	1 (358.4, 446.8)	2 (418.2, 3401.2)

Notes. 1. Firms' performance measures:

$\Delta EPS_{(t,t+1)}^f$ is a firm's EPS growth rate at year $t + 1$ relative to year t . Formally,

$$\Delta EPS_{(t,t+1)}^f = (EPS_{t+1}^f - EPS_t^f) / |EPS_t^f|$$

$SAR_{(t+1,t+2)}^f$ is the cumulative return from April of year $t + 1$ to March of year $t + 2$, minus the return of the corresponding market decile over the same period.

2. SVM-score and SVM-prob models are built on three binary SVM classifiers for the three classes of top-25%, middle-50%, and bottom-25% firms. Specifically, the SVM-score model uses the highest score from the three binary classifiers to assign a class label to a data point. The SVM-prob model transforms the scores of the three binary classifiers into probabilities using the Lin–Platts method, and then uses the highest probability to assign a class label to a data point. The SVM-multi model is the multiclass classification package implemented by Joachims based on Crammer and Singer's (2002) study.

3. The first row of cell entries represents each classification model's average relative rank in predictive accuracy. A smaller number represents better rank among the three models. Values in parentheses indicate the mean accuracy value and its variance. Bolded entry is the top-performing classification model in accuracy.

4. The second row of cell entries represents each classification model's average relative rank in predictive cost. A small number represents better rank among the three models. Values in parentheses indicate the mean cost and its variance. The bolded entry is the top-performing classification model in predictive cost. Cost is defined as:

$$Cost = \begin{cases} 2 \times C, & \text{if } \begin{cases} \text{TrueClass} = \text{outperform, and} \\ \text{PredictedClass} = \text{underperform or} \\ \text{TrueClass} = \text{underperform, and} \\ \text{PredictedClass} = \text{outperform} \end{cases} \\ C, & \text{otherwise} \end{cases}$$

RQ1: What are the relative merits of the different classifiers when applied to our goal? (Here, we compare the different SVM-based classifier models.)

Table 2 shows the average ranking of the models by predictive accuracy and by error cost with this design. We observe that SVM-prob ranks the best on average in accuracy for both EPS and SAR financial measures. SVM-prob ranks the best on average in cost with SAR and comes in as a very close second after SVM-multi. Overall, SVM-prob stands out as the best in terms of both predictive accuracy and cost of errors.

We continued to explore differences between the three SVM classifiers in terms of types of errors made. Table 3 presents the classification contingency tables of SVM-score, SVM-prob, and SVM-multi for predicting $SAR_{(t+1,t+2)}^f$ averaged over years. SVM-prob was not able to predict outperforming firms with SAR, but the other two models were able to do so. Another interesting observation is that in classifying outperforming and underperforming firms, SVM-score and SVM-multi appear to have different patterns. SVM-score's correct prediction for outperforming and underperforming is higher in percentage terms than is SVM-multi (17.96 vs. 8.86% and 24.33 vs. 16%, respectively). On the other hand, SVM-score's error rates of misclassifying outperforming as underperforming and vice versa also are higher than is SVM-multi (20.5 vs. 15.07% and 17.52 vs. 10.59%, respectively). These observations stayed consistent in predicting change in EPS. We observe that SVM-score risks higher cost of errors for achieving higher predictive accuracies when

predicting outperforming and underperforming classes. SVM-multi, on the other hand, achieved lower error rates at the cost of lower predictive accuracies when predicting the outperforming and underperforming classes. However, neither of these two models provided overall accuracies better than that of the SVM-prob model, although the latter failed to predict outperforming firms.

We further analyzed the SVM-prob model's performance against the majority vote baseline. The majority vote baseline accuracy is 50%. Figure 4 shows that SVM-prob offers the same or better accuracy as compared to baseline for 9 of 10 measure/year combinations. Of the nine measure/year combinations, the model was able to predict better than baseline for six measure/year combinations while it predicted the same as baseline in three measure/year combinations. Of the six better-than-baseline combinations, three are for $\Delta EPS_{(t,t+1)}^f$, and three are for $SAR_{(t+1,t+2)}^f$. Note that with this predictor, we build three one-against-rest SVM classifiers—one for each class. When a given instance is being classified, we transform its three SVM scores into probabilities using Lin–Platt's method (Lin et al., 2003). We then assign the instance to the class with the highest probability.

Considering that our experiment design reflects real-world application where we use a model trained with one year of data to predict for the following year, we would not be able to apply a test for statistical significance to our model performance as we could with a standard cross-validation design. Our design does not assume perfect

TABLE 3. Contingency table: three SVM models for predicting $SAR_{(t+1,t+2)}^f$.

25–50–25% Class definition	SVM-score prediction			%True distribution
	%Outperform	%Average	%Underperform	
True outperform	17.96	64.52	17.52	25.08
True average	18.96	64.21	16.83	49.99
True underperform	20/50	55.17	24.33	24.94
Prediction distribution	19.09	62.04	18.87	100

25–50–25% Class definition	SVM-prob prediction			%True distribution
	%Outperform	%Average	%Underperform	
True outperform	0	94.36	5.64	25.08
True average	0	94.09	5.91	49.99
True underperform	0	85.53	14.47	24.94
Prediction distribution	0	92.03	7.97	100

25–50–25% Class definition	SVM-multi prediction			%True distribution
	%Outperform	%Average	%Underperform	
True outperform	8.86	80.54	10.59	25.08
True average	8.83	81.90	9.27	49.99
True underperform	15.07	68.93	16.00	24.94
Prediction distribution	10.40	78.33	11.28	100

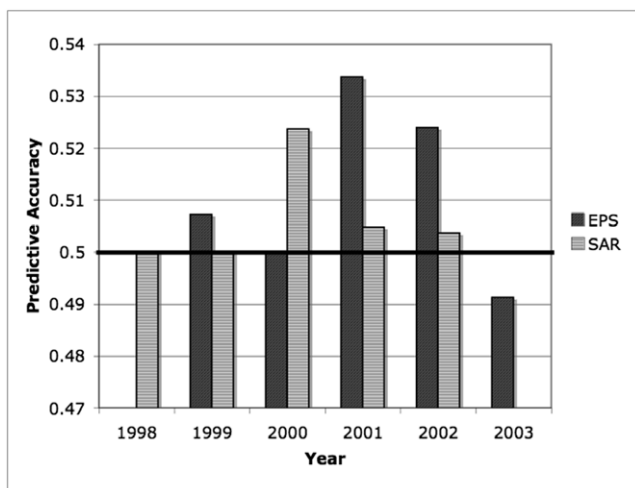


FIG. 4. SVM-prob accuracy compared with majority vote baseline.

knowledge of the general economic climate for each year, which critically raises the difficulty level of our experiment. Considering the apparently marginal improvement of our model's performance over the majority vote baseline in predicting SAR, we employ the kappa statistic, a measure to evaluate interrater reliability, to test the agreement between our model prediction for SAR and the true class labels of the test data points. We obtained 0.0509 for kappa and 0.0670 for weighted kappa, indicating our model's significant agreement with the test data's true distribution at the 0.01% level. Although we are pleased with the performance of our classification models over the majority vote baseline, we

extend our evaluation with respect to other benchmarks, as discussed next. The regression performance also will be included in the discussion on our second research question.

We will now address the second and third research questions together.

RQ2: How do our different supervised learning models perform compared to analysts' forecasts? (Here, we test our best SVM-based classifier and our regression model.)

RQ3: Are annual reports suitable for predicting change in certain financial measures? (Here, we compare EPS and SAR.)

Figure 5 compares the accuracy of analyst forecasts and that of our SVM-prob model in terms of predicting EPS. It is clear that analysts are consistently more accurate than is our SVM-prob model. Analysts offer improvements of at most 20%. This result coincides with other findings that have suggested that "analysts have consistently been shown to forecast earnings more accurately than do mechanical models" (Schipper, 1991, p. 107). The reason is likely that analysts "have access to more information to project future earnings than the accounting system has to produce the earnings number" (Schipper, 1991, p. 107).

Interestingly, the comparison with analysts for $SAR_{(t+1,t+2)}^f$ yields very different results. Figure 6 compares accuracies of analyst recommendations with accuracies of our SAR predictors. Analyst predictions were considerably less accurate than was our SVM-prob model. Analyst accuracies also were lower than the majority vote baseline for all years studied. The large difference between the analysts' stock forecast and the majority vote baseline is

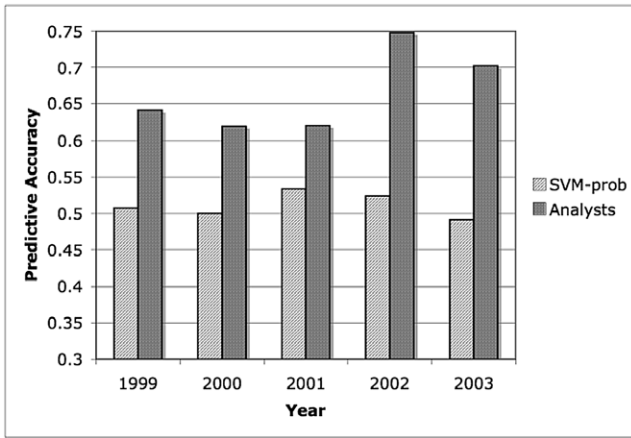


FIG. 5. Comparing accuracy for predicting $\Delta EPS_{(t,t+1)}^f$: SVM-prob versus analysts' forecasts.

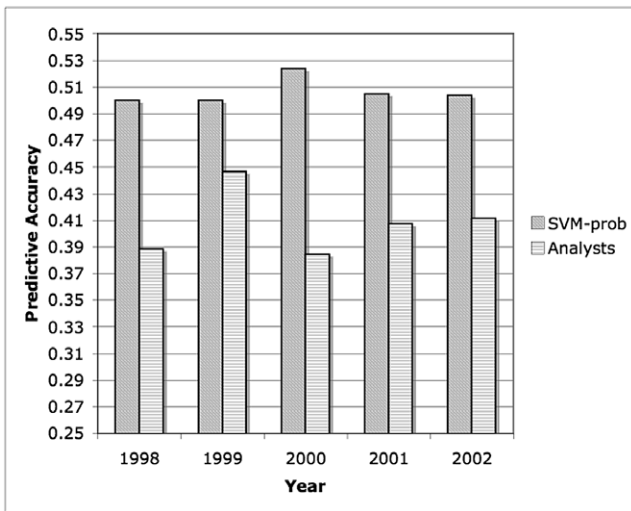


FIG. 6. Comparing accuracy for predicting $SAR_{(t+1,t+2)}^f$: SVM-prob versus analysts' forecasts.

consistent with findings from other studies that analyst recommendations are biased because of the economic incentives faced by sell-side brokerage firms (Jegadeesh, Kim, Krische, & Lee, 2004). Our model does not exhibit such a bias. Future studies could combine our model's predictions with analysts' forecasts to yield stronger results, which in turn may shed further light on the nature of the analysts' information source.

The observations about how predictive models compare with analysts' forecast are further validated with the results from regression models. As seen in Tables 4 and 5, using Kendall's $\hat{\tau}$ and Spearman's $\hat{\rho}$, the analysts' forecast for $\Delta EPS_{(t,t+1)}^f$ were outstanding not only against the regression model but also significantly better than were analysts' forecasts for SAR. On the other hand, although regression models were clearly ineffective in predicting $\Delta EPS_{(t,t+1)}^f$, we are able to achieve much better performance than did the

analysts in predicting $SAR_{(t+1,t+2)}^f$ in 4 of the 5 years. These findings further confirm our observations as well as speculations of researchers in the accounting domain that (a) analysts probably have access to additional valuable information resources in predicting $\Delta EPS_{(t,t+1)}^f$, and (b) analysts may be biased in their stock recommendations due to managerial incentives. Automatic supervised learning models may be more adept at predicting long-term market response than a firm's profitability.

RQ4: How do our predictive models perform over different years? (Here, we compare experiment results with data around the financially turbulent year of 2000.)

We look into our models' performance trend by year in predicting $SAR_{(t+1,t+2)}^f$ since this is where our models are most effective. Figure 7 presents how our best classifier model (i.e., SVM-prob) as well as analysts perform in different years. Apart from the fact that our classifier outperformed analysts in all years, it performed the best for Year 2000, which interestingly was the hardest year to predict for analysts. In Figure 8, the blue and yellow lines show how our regression model performed in different years, while the red and green lines indicate the analysts' performance. Although the regression model outperformed the analysts in 4 of 5 years, the hardest year for the regression model and the only year it failed to compete with analysts is, interestingly, Year 2000, the easiest year to predict for our classifier model. We conclude that regarding the different learning models (classifier vs. regression), the general predictive trend by years for our supervised models is consistent when compared with that of analysts. However, the classification model and the regression model perform differently in terms of the hardest and the easiest years to predict. What may seem a hard year to predict for regression turned out to be the easiest year for the classifier model. This difference probably comes from the fundamental difference in the underlying learning mechanisms defining these two supervised learning models, respectively. What is more interesting is that this observation sheds light on the importance of choosing the right prediction question (i.e., a coarser categorization of performance vs. a finer and more precise real value prediction) and consequently the right corresponding predictive models (i.e., classifier vs. regression), given the same data set spanning different time periods.

Conclusions

Mandatory company annual reports, one of the most important sources of disclosures, have remained largely untapped by the machine learning and text mining community. Previous studies have shown that the narrative discussions in annual reports are important for assessing firm performance. We conducted research to systematically study the potential of building models to predict firm performance from their annual reports using supervised learning methods. We presented results from research structured along five dimensions: learning models, financial measures, experiment design, document representations, and evaluation measures. Using two types of supervised learning

TABLE 4. Evaluating regression models and analysts' forecasts for predicting $\Delta EPS_{(t,t+1)}^f$

Year	Kendall's $\hat{\tau}$		Spearman's $\hat{\rho}$	
	Regression model	Analysts' forecasts	Regression model	Analysts' forecasts
1999	0.0026	0.3932	−0.0033	0.4851
2000	0.0649	0.4551	0.0910	0.5860
2001	−0.1091	0.4222	−0.1552	0.5348
2002	0.0134	0.5573	0.0154	0.6863
2003	−0.0192	0.5033	−0.0264	0.6259

Note. Cell entries are the values of Kendall's $\hat{\tau}$ and Spearman's $\hat{\rho}$, the nonparametric correlation based on sample quantities, for the regression model and the analysts' forecasts in predicting $\Delta EPS_{(t,t+1)}^f$. Formally, $\hat{\tau} = 1 - \frac{4T}{n(n-1)}$; $\hat{\rho} = 1 - \frac{12R}{n(n-1)(n+1)}$. The range of these two measures is $[-1, 1]$, where 1 corresponds to perfect model performance, and -1 corresponds to making all possible errors. The bolded entries indicate the winning model's performance.

TABLE 5. Evaluating the regression models and analysts' forecasts for predicting $SAR_{(t+1,t+2)}^f$

Year	Kendall's $\hat{\tau}$		Spearman's $\hat{\rho}$	
	Regression model	Analysts' forecasts	Regression model	Analysts' forecasts
1998	0.0161	0.0080	0.0226	−0.0167
1999	0.1416	−0.0083	0.2144	−0.0331
2000	−0.3152	0.1680	−0.4635	0.2304
2001	0.1753	0.1015	0.2609	0.1296
2002	0.1451	0.0944	0.2205	0.1177

Note. Cell entries are the values of Kendall's $\hat{\tau}$ and Spearman's $\hat{\rho}$, the nonparametric correlation based on sample quantities, for the regression model and the analysts' forecasts for predicting $SAR_{(t+1,t+2)}^f$. Formally, $\hat{\tau} = 1 - \frac{4T}{n(n-1)}$; $\hat{\rho} = 1 - \frac{12R}{n(n-1)(n+1)}$. The range of these two measures is $[-1, 1]$, where 1 corresponds to perfect model performance, and -1 corresponds to making all possible errors. The bolded entries indicate the winning model's performance.

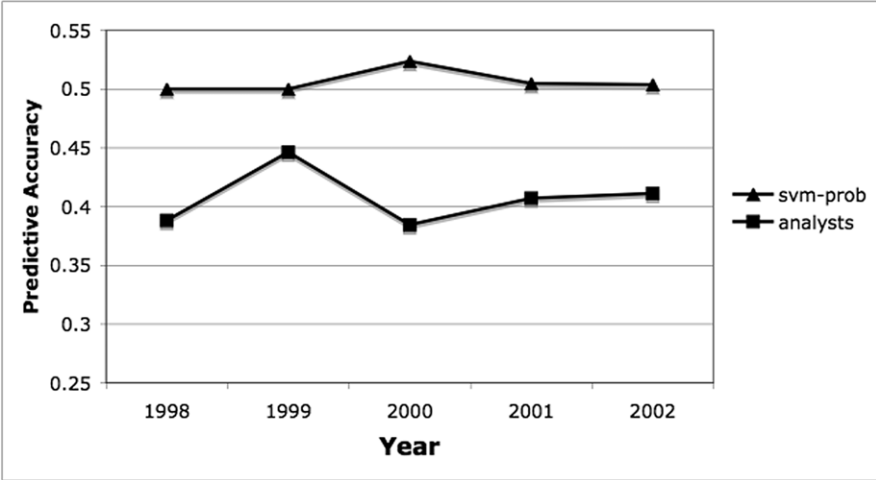


FIG. 7. Evaluating the classifier model and analysts' forecasts over different years.

models (i.e., classification and regression), we address four research questions raised to study from different angles the predictive potential of supervised learning models built with annual reports. The results we obtain are quite encouraging in that we not only identify the different models' strengths and weaknesses but our models achieve better performance than do analysts' forecasts in predicting size-adjusted

cumulative return. Our major findings in evaluating models against analysts' forecasts provide further evidence regarding speculations about analysts' information resources and possible bias in their prediction. We have conducted a much wider range of experiments than are presented here. Strategies were not presented that produced weaker results (e.g., return on equity as finan-

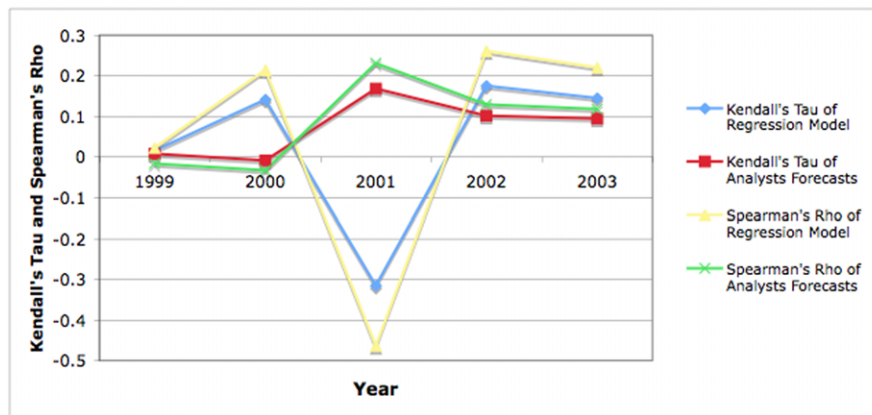


FIG. 8. Evaluating the regression models and analysts' forecasts over different years. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

cial indicator) or explorative tests (e.g., 10-fold cross-validation tests for all indicators and across all years). In response to suggestions by reviewers, we also tested a different design where the training data used to build the classifier or to regress include all the years previous to the test year. That is, we use the data corresponding to years $t - 3$, $t - 2$, and $t - 1$ to predict year t . We tested these for both EPS and SAR measures. The results when compared with those of the single-year training data model presented in the article show similar patterns in terms of performance against our benchmark and in terms of the years most difficult or easiest to predict. Since we did not observe significantly different results, we only document detailed discussion of the current single test-year model in this article.

Our findings should be relevant to academics interested in such issues as evaluating information-disclosure quality and in methods for handling text mining problems in the business domain. The results of our analysis also should be of interest to policymakers in considering information efficiency in capital markets, and to general investors in using market information and analysts' forecasts to assist their investment decisions.

Our study is limited to annual reports as a form of mandatory disclosure. While the supervised-learning framework is adaptable to include other information resources such as quarterly report, we could consider improving our model's performance by incorporating other elements such as quantitative variables, economic forecasts, analyst forecasts, and industry and product life cycle. Further, it would be of interest to compare the analysis of mandatory disclosure with voluntary disclosure and third-party disclosure. In conclusion, we see considerable scope for future research along this direction.

Acknowledgments

We thank Professor Wang Huai Qing, Dr. Rong Zheng, and Michael Rechenthin for their valuable suggestions and

thoughtful feedback. This work was partially supported by Grants 61070061, 71271061, and 70801020 from the National Natural Science Foundation, China, Grant 12YJAH103 from the Ministry of Education of China Project, and Grant 2011J5100004 from the Guangzhou Science and Technology Program Project.

References

- Balakrishnan, R., Qiu, X.Y., & Srinivasan, P. (2010). On the predictive ability of narrative disclosures in annual reports. *European Journal of Operational Research*, 202(3), 789–801.
- Chu, W., & Keerthi, S. (2007). Support vector ordinal regression. *Neural Computation*, 19(3), 792–815.
- Crammer, K., & Singer, Y. (2002). On the algorithm implementation of multiclass SVMs. *Journal of Machine Learning Research*, 2, 265–292.
- Das, S.R., & Chen, M.Y. (2007). Yahoo! for Amazon: Sentiment extraction from small talk on the web. *Management Science*, 53(9), 1375–1388.
- Feldman, R., Govindaraj, S., Livnat, J., & Segal, B. (2010). Management's tone change, post earnings announcement drift and accruals. *Review of Accounting Studies*, 15(4), 915–953.
- Fields, T.D., Lys, T.Z., & Vincent, L. (2001). Empirical research on accounting choice. *Journal of Accounting and Economics*, 31, 255–307.
- Gunning, R. (1952). *The technique of clear writing*. New York: McGraw-Hill.
- Jegadeesh, N., Kim, J., Krusche, S.D., & Lee, C.M.C. (2004). Analyzing the analysts: When do recommendations add value? *Journal of Finance*, 59(3), 1083–1124.
- Joachims, T. (1998). Text categorization with support vector machines: Learning with many relevant features. In C. Nedellec & C. Rouveirol (Eds.), *Lecture notes in computer science: Vol. 1398*. In *Proceedings of the 10th European Conference on Machine Learning (ECML-98)*, Chemnitz, Germany (pp. 137–142). Berlin, Germany: Springer.
- Kloptchenko, A., Eklund, T., Back, B., Karlsson, J., Vanharanta, H., & Visa, A. (2004). Combining data and text mining techniques for analyzing financial reports. *Intelligent Systems in Accounting, Finance and Management*, 12(1), 29–41.
- Kotsiantis, S.B. (2007). Supervised machine learning: A review of classification techniques. In *Proceedings of the Conference on Emerging Artificial Intelligence Applications in Computer Engineering* (pp. 3–24). Amsterdam, The Netherlands: IOS Press.
- Lehavy, R., Li, F., & Merkley, K. (2011). The effect of annual report readability on analyst following and the properties of their earnings forecasts. *The Accounting Review*, 86(3), 1087–1115.

- Li, F. (2008). Annual report readability, current earnings, and earnings persistence. *Journal of Accounting and Economics*, 45(2–3), 221–247.
- Li, F. (2010). The information content of forward-looking statements in corporate filings—A naïve Bayesian machine learning approach. *Journal of Accounting Research*, 48(5), 1049–1102.
- Li, Y., & Bontcheva, K. (2008). Adapting support vector machines for F-term-based classification of patents. *ACM Transactions on Asian Language Information Processing*, 7(2). Retrieved from <http://doi.acm.org/10.1145/1362782.1362786>
- Li, Y., Gond, S., & Liddell, H. (2000). Support vector regression and classification based multi-view face detection and recognition. *Proceedings of the Fourth Institute of Electrical and Electronic Engineers (IEEE), pp. 300–305* International Conference on Automatic Face and Gesture Recognition.
- Lin, H.T., Lin, C.J., & Weng, C.H. (2007). A note on Platt's probabilistic outputs for support vector machines. *Machine Learning*, 68(3), 267–276.
- Magnusson, C., Arppe, A., Eklund, T., Barbro, B., Vanharanta, H., & Visa, A. (2005). The language of quarterly reports as an indicator of change in the company's financial status. *Information & Management*, 42(4), 561–574.
- Manning, C., & Raghavan, P. (2008). *Introduction to information retrieval*. New York, NY: Cambridge University Press.
- Mittermayer, M. (2004). Forecasting intraday stock price trends with text mining techniques. *Proceedings of the Hawaii International Conference on System Sciences*. doi:10.1109/HICSS.2004.1265201
- Moschitti, A., & Basili, R. (2004). Complex linguistic features for text classification: A comprehensive study. In S. McDonald and J. Tait (Eds.), *ECIR 2004, LNCS 2997* (pp. 181–196). Heidelberg, Germany: Springer-Verlag.
- Platt, J.C. (1999). Probabilistic outputs for support vector machines and comparison to regularized likelihood methods. *Advances in large margin classifiers* (pp. 61–74). Cambridge, MA: MIT Press.
- Qiu, X.Y. (2010). Towards building ranking models with annual reports. *Journal of Digital Information Management*, 8(5), 338–343.
- Qiu, X.Y., Srinivasan, P., & Street, N. (2006). Exploring the forecasting potential of company annual reports. *Proceedings of the American Society for Information Science and Technology*, 43(1), 1–15.
- Rice, S.B., Nenadic, G., & Stapley, B.J. (2005). Mining protein function from text using term-based support vector machines. *BMC Bioinformatics*, 6(Suppl. 1), S22.
- Robertson, C., Geva, S., & Wolf, R.C. (2007). Can the content of public news be used to forecast abnormal stock market behavior? In *Proceedings of the Seventh Institute of Electrical and Electronic Engineers (IEEE) International Conference on Data Mining* (pp. 637–642).
- Rosset, S., Perlich, C., & Zadrozny, B. (2007). Ranking-based evaluation of regression models. *Knowledge and Information Systems*, 12(3), 331–353.
- Saad, E.W., Prokhorov, D.V., & Wunsch, D.C. (1998). Comparative study of stock trend prediction using time delay, recurrent and probabilistic neural networks. *IEEE Transaction on Neural Networks*, 9(6), 1456–1470.
- Schipper, K. (1991). Analysts' forecasts. *Accounting Horizons*, 5(4), 105–121.
- Schumaker, R.P., & Chen, H. (2009). Textual analysis of stock market prediction using breaking financial news: The AZFintext system. *ACM Transactions on Information Systems*, 27(12). doi:10.1145/1462198.1462204
- Sebastiani, F. (2002). Machine learning in automated text categorization. *ACM Computing Surveys*, 34(1), 1–47.
- Shatkay, H., & Feldman, R. (2003). Mining the biomedical literature in the genomic era: An overview. *Journal of Computational Biology*, 10(6), 821–855.
- Shatkay, H., Pan, F., Rzhetsky, A., & Wilbur, W.J. (2008). Multi-dimensional classification of biomedical text: Toward automated, practical provision of high-utility text to diverse users. *Bioinformatics*, 24(18), 2086–2093.
- Smith, M., & Taffler, R.J. (2000). The chairman's statement: A content analysis of discretionary narrative disclosures. *Accounting, Auditing & Accountability Journal*, 13(5), 624–646.
- Smola, A.J., & Scholkopf, B. (2004). A tutorial on support vector regression. *Statistics and Computing*, 14(3), 199–222.
- Srinivasan, P., & Qiu, X.Y. (2007). Go for gene documents. *BMC Bioinformatics*, 8(Suppl. 9), S3.
- Steel, R.G.D., & Torrie, J.H. (1960). *Principles and procedures of statistics with special reference to the biological sciences*. New York, NY: McGraw Hill.
- Tetlock, P.C., Saar-Tsechansky, M., & Macskassy, S. (2008). More than words: Quantifying language to measure firms' fundamentals. *Journal of Finance*, 63(3), 1437–1467.
- Wu, C.-H., Ho, J.-M., & Lee, D.T. (2004). Travel-time prediction with support vector regression. *IEEE Transactions on Intelligent Transportation Systems*, 5(4), 276–281.
- Zhai, Y., Hsu, A., & Halgamuge, S.K. (2007). Combining news and technical indicators in daily stock price trends prediction. In *Proceedings of the Fourth International Symposium on Neural Networks: Advances in Neural Networks, Part III* (pp. 1087–1096).
- Zhang, W., Cao, Q., & Schniederjans, M. (2004). Neural network earnings per share forecasting models: A comparative analysis of alternative methods. *Decision Sciences*, 35(2), 205–237.